

RF Architecture, Protocol Analysis, and Reverse-Engineering Methodology for the Retekess TM102 Wireless Buzzer System (FCC ID 2A3NOTM101)

1.0 Introduction and Scope of Analysis

The proliferation of proprietary wireless communication systems in consumer electronics necessitates rigorous physical layer and protocol-level analyses to ensure security, interoperability, and spectrum compliance. This report executes an exhaustive technical evaluation of the Retekess TM101 and TM102 wireless game buzzer systems, operating under the Federal Communications Commission (FCC) identifier 2A3NOTM101. Manufactured by Zhengzhou Ysair Technology Co., Ltd. and distributed through Henan Eshow Electronic Commerce Co., Ltd., the system is designed to facilitate high-speed, multi-node interactive polling and trivia events.

The physical hardware architecture is bifurcated into two primary operational classes: a central coordinating host (the controller or transmitter) and up to 32 remote edge nodes (the responders or buzzers). Physical examination of the hardware labels reveals that both the host controller and the remote responders operate under the same FCC ID (2A3NOTM101), indicating highly symmetrical internal RF circuitry despite their disparate functional roles [cite: image 1, image 3]. The host controller requires a 5V, 1A input via a USB Type-C interface, while the remote responder edge nodes are powered by three 1.5V AAA batteries [cite: 3, image 1, image 3]. The system is rated for an 80-meter transmission distance in open-air environments. The primary objective of this documentation is to synthesize publicly available FCC Office of Engineering and Technology (OET) database exhibits—specifically focusing on Operational Descriptions, Block Diagrams, and RF Test Reports compiled by the Shenzhen LCS Compliance Testing Laboratory—to reconstruct the system's underlying communication mechanics. By systematically mapping the network topology, component duplexing capabilities, radio frequency characteristics, and protocol framing timing, this report provides a comprehensive foundation for signals intelligence (SIGINT) capture and baseband reverse-engineering.

2.0 System Architecture and Network Topology

The foundational element of any multi-node wireless ecosystem is its network topology, which defines how data propagates across the physical medium and how the system resolves simultaneous transmissions. The Retekess TM102 system is engineered to function as a "fastest finger first" responder, necessitating extreme precision in recording the sequential order of button presses from up to 32 independent actors. This requirement fundamentally dictates the architectural paradigms the hardware must employ.

2.1 The Case Against Synchronous Polling

A pivotal question in reverse-engineering this system is determining whether the central controller utilizes a synchronous polling mechanism to interrogate each buzzer, or if the edge nodes operate asynchronously. A thorough evaluation of the hardware constraints, latency requirements, and power profiles definitively refutes the use of synchronous polling. In a traditional polling architecture, the host controller would iterate through a predefined list of node addresses (e.g., Buzzer 1 through Buzzer 32), transmitting a query packet and waiting for a response or a timeout before moving to the next node. While this eliminates airwave collisions, it introduces catastrophic latency scaling. Assuming a highly aggressive polling cycle where a query and response take a combined two milliseconds, a complete round-robin interrogation of 32 nodes would consume 64 milliseconds. In a competitive game scenario, a player pressing Buzzer 32 could physically actuate their button tens of milliseconds before the player on Buzzer 1, but because Buzzer 1 is queried earlier in the cycle, the system would incorrectly award the win to Buzzer 1. This non-deterministic latency destroys the integrity of the arbitration process.

Furthermore, continuous polling requires the remote buzzers to maintain their transceiver ICs in a constant active-receive (RX) state to listen for their specific interrogation packet. Receiving RF data consumes significant current—often equal to or greater than transmission current in low-cost 2.4 GHz application-specific integrated circuits (ASICs). Maintaining an active RX window would rapidly drain the three AAA batteries powering the responder [cite: image 3]. The manufacturer documentation asserts that the button can operate for up to 25 hours on a single charge or battery cycle, which is mathematically incompatible with a continuous 100% RX duty cycle.

2.2 Asynchronous Interrupt Architecture

The FCC Operational Description and the physical constraints of the hardware heavily support an asynchronous, interrupt-driven architecture. In this paradigm, the remote buzzers spend the vast majority of their operational lifecycle in an ultra-low-power deep sleep state, consuming only microamperes of current.

The physical buzzer mechanism, engineered to withstand over 20,000 compression tests, is directly tied to a hardware interrupt pin on the internal microcontroller (MCU). When a player presses the button, the mechanical switch closes a circuit, triggering an instant edge-detected interrupt. Within microseconds, the MCU wakes, initializes the RF phase-locked loop (PLL), and fires an asynchronous burst packet into the RF medium indicating its state change. This topology is strictly point-to-multipoint (a star network), where all edge nodes communicate exclusively and directly with the central arbiter, and never with each other.

2.3 Collision Domains and Arbitration Algorithms

Because the remote nodes transmit asynchronously upon human actuation, the system inherently operates within a highly contested collision domain. If multiple players perceive a stimulus simultaneously, their respective buzzers will wake and transmit their interrupt packets concurrently. Without mitigation, these RF waves superimpose in the physical medium, causing destructive interference at the host's receiver antenna, resulting in a loss of all data.

To resolve the "winning" buzzer with microsecond precision, the system must employ robust collision resolution mechanics. Based on the operational behaviors described in the product

literature, the system likely implements a proprietary variant of Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) combined with aggressive randomized backoff timers. Upon actuating a transmission, a buzzer immediately switches its transceiver to receive mode to await a deterministic acknowledgment (ACK) from the host controller. The host controller's MCU serves as the ultimate temporal arbiter; it records the precise arrival timestamp of the first valid, uncorrupted packet it successfully demodulates. The host then immediately broadcasts a targeted ACK declaring that specific node the winner. If two buzzers transmit simultaneously and collide, neither receives an ACK within the strict timeout window. Both edge nodes then generate a pseudo-random integer to determine a microsecond-scale backoff duration before attempting a retransmission. The host continues to record subsequent valid packets to determine the second, third, fourth, and fifth-place responders, as evidenced by the system's ability to rank the first five presses.

Topology Characteristic	Technical Implementation	Implication for Reverse Engineering
Architecture	Point-to-Multipoint (Star)	Capture devices only need to monitor the host's operating channel; mesh hopping is absent.
Node State	Asynchronous Interrupt	Signals will appear as highly sparse, unpredictable microsecond bursts. Continuous recording is inefficient.
Collision Handling	Randomized Backoff / ACK	The capture will show multiple rapid retries from the same node if an ACK is not received, requiring careful payload analysis to avoid duplicate counting.

3.0 Transceiver Capabilities and Duplexing Mechanics

A core requirement for tuning an RF capture device and mapping the communication flow is determining the bidirectional nature of the radio links. The question of whether the remote edge nodes are simplex (transmit-only) devices or half-duplex transceivers can be resolved through an analysis of the system's supported interaction paradigms and the test report documentation from the Shenzhen LCS Compliance Testing Laboratory.

3.1 Refutation of Simplex Operation

Simplex RF systems, such as basic environmental temperature sensors or legacy garage door openers, feature a physical layer capable only of one-way data propagation. If the TM102 responders were simplex devices, they would transmit their button-press interrupt to the host but possess no hardware capability to receive data back.

The physical functionality of the TM101/TM102 system conclusively invalidates the possibility of simplex operation. The system relies on intricate, host-driven visual feedback mechanisms that necessitate active downlink data reception at the edge nodes.

3.2 Half-Duplex State Machines and Visual Feedback

The remote buzzers are equipped with multi-colored LED arrays designed to visually communicate the arbitration results to the players in real time. The color sequencing is highly specific and dynamically controlled by the central host. According to the manufacturer's operational manual, the first buzzer to successfully register a press illuminates with a steady green light, the second buzzer flashes green, the third buzzer illuminates steady blue, the fourth flashes blue, and the fifth flashes yellow.

Because the edge nodes cannot communicate with one another to determine their relative ranking, this state must be calculated by the host controller and broadcast back to the specific buzzers. This definitively proves that the buzzers are equipped with half-duplex RF transceivers. Furthermore, the controller interface features specialized administrative functions that require the buzzers to process command packets. The host interface (visualized in the provided documentation) includes a "Lock" function, a "Clear" function, and buttons to switch between "General", "Rule", and "Random" modes [cite: image 2].

- In **Rule Mode**, the game mechanics dictate that players are prohibited from answering until the buzzer light transitions to a steady blue state. If a player presses the button while it is flashing, the system registers a foul, transitions the LED to red, and locks the buzzer out of the current round.
- In **Random Mode**, the host initiates a sequence where all buzzer lights flash simultaneously; the host then transmits a stop command, causing one randomly selected buzzer to illuminate green while all others lock and become unresponsive.

These operational modes require the buzzers to actively listen for state-change packets from the host. Therefore, the buzzers transition from their asynchronous interrupt state into a highly synchronized listening state when commanded by the host, maintaining their receivers in an active configuration during specific game phases.

3.3 Internal Hardware Architecture Deductions

The confirmation of half-duplex operation provides deep insights into the internal hardware block diagrams submitted for FCC certification. The internal architecture of the responder must consist of a microcontroller (likely an 8-bit or 32-bit ARM Cortex-M0 derivative) interfaced via a Serial Peripheral Interface (SPI) to a discrete 2.4 GHz RF transceiver ASIC, or a highly integrated System-on-Chip (SoC) that combines both functions.

The MCU handles the debounce logic of the physical switch, manages the RGB LED pulse-width modulation (PWM) drivers for color mixing, and constructs the data payloads. The RF ASIC handles the Gaussian filtering, modulation, and physical transmission of the data.

Because both the host and the buzzers share the identical FCC ID 2A3NOTM101 [cite: image 1, image 3], it is highly probable that both components utilize the exact same RF ASIC and underlying printed circuit board (PCB) RF layout, merely loaded with different firmware configurations to dictate their respective roles as coordinator or edge node.

Component Role	Duplex Capability	Primary TX Function	Primary RX Function
Host Controller	Half-Duplex	Broadcast ACKs, State Changes (Lock/Clear), Mode Commands	Receive asynchronous interrupts from up to 32 edge nodes.
Remote Buzzer	Half-Duplex	Transmit hardware interrupt payloads, battery status	Receive ACK rankings, LED color commands, mode synchronization.

4.0 Physical Layer (PHY) and Radio Frequency Characteristics

To effectively isolate and capture the proprietary wireless signals utilizing an ESP32 and nRF24L01+ sniffer setup, the exact physical layer parameters must be extracted from the FCC test reports and physical labels. The Shenzhen LCS Compliance Testing Laboratory evaluated the 2A3NOTM101 device for compliance with regulatory emissions standards.

4.1 Frequency Allocation and Band Edge Compliance

The physical label affixed to the underside of the responder clearly denotes an operating frequency range of 2.405 GHz to 2.485 GHz [cite: image 3]. This allocation places the system within the universally adopted 2.4 GHz Industrial, Scientific, and Medical (ISM) band, which spans from 2.400 GHz to 2.4835 GHz.

However, a critical discrepancy exists between the physical label marketing and the empirical engineering data filed with the FCC. The official test reports and User Manual technical specifications submitted to the OET database document a highly specific operational frequency range of **2420 MHz to 2465 MHz**.

This nuance reveals a vital third-order insight regarding the firmware engineering of the device. While the silicon of the integrated RF transceiver is physically capable of tuning across the entire 2.4 GHz ISM band (from 2.405 to 2.485 GHz), the firmware engineers have artificially restricted the operational channel map to between 2420 MHz and 2465 MHz. This spectral padding is a common regulatory compliance strategy. By avoiding the absolute edges of the ISM band (2400 MHz and 2483.5 MHz), the manufacturer ensures that sideband splatter and harmonic phase noise do not bleed into adjacent restricted spectrums, guaranteeing passage

of strict FCC band-edge emissions tests. For the reverse engineer, this means the ESP32 channel-hopping sweep parameters should be strictly confined to the 2420-2465 MHz window, ignoring the wider range printed on the chassis.

4.2 Modulation Physics: The Role of GFSK

The physical label and FCC filings explicitly state that the system utilizes **Gaussian Frequency Shift Keying (GFSK)** modulation [cite: image 3, 45, 60, 96]. Understanding the physics of GFSK is critical for configuring the nRF24L01+ radio module.

Standard Frequency Shift Keying (FSK) encodes digital data by shifting the carrier frequency between two discrete values—representing binary 1 and 0. However, the instantaneous switching between these frequencies creates sharp phase transitions in the time domain, which mathematically translates to extreme high-frequency noise in the frequency domain. This phenomenon, known as spectral splatter, causes the signal to consume an unnecessarily wide bandwidth, causing severe adjacent-channel interference.

GFSK mitigates this by passing the baseband digital data stream through a Gaussian low-pass filter before it enters the voltage-controlled oscillator (VCO). This filter smooths the square wave pulses, rounding off the sharp transitions. The result is a highly continuous phase shift that drastically narrows the spectral footprint of the transmission. For a system designed to support 32 independent nodes in a crowded 2.4 GHz environment, GFSK is functionally mandatory to maintain spectral efficiency and reliable link margins.

4.3 Channel Bandwidth and Spectral Efficiency

While the specific channel bandwidth is buried deep within the test report emission masks, proprietary GFSK ASICs utilized in low-cost consumer electronics (such as the ubiquitous Nordic Semiconductor nRF24 series, the Beken BK2425, or the Panchip XN297L) universally standardize around **1 MHz or 2 MHz channel bandwidths**.

Given the operational window of 2420 MHz to 2465 MHz, a 1 MHz bandwidth would allow for up to 45 discrete, non-overlapping channels. The host controller likely performs a passive energy detect (clear channel assessment) upon boot, scanning this 45 MHz swath to identify the channel with the lowest ambient noise floor (avoiding local Wi-Fi and Bluetooth interference), and then commands the paired buzzers to rendezvous on that specific center frequency.

4.4 Transmission Power, Link Budget, and Free Space Path Loss

The manufacturer claims a robust operational range of up to 80 meters (262 feet) in open space environments. Operating at 2.4 GHz, achieving this distance without the aid of highly directional, high-gain antennas requires a carefully balanced link budget.

The theoretical attenuation of the RF signal as it propagates through an open environment can be modeled using the Free Space Path Loss (FSPL) equation:

$$\text{FSPL (dB)} = 20 \log_{10}(d) + 20 \log_{10}(f) - 147.55$$

Where d is the distance in meters (80) and f is the center frequency in Hertz (2.44×10^9). Executing this calculation yields a path loss of approximately 78.2 dB.

Standard integrated PCB trace antennas (commonly inverted-F or meander line designs) typically feature a gain of 0 dBi to +2 dBi. The receiver sensitivity (noise floor) for low-cost GFSK ASICs operating at a 1 Mbps signaling rate is generally rated at -90 dBm. To maintain a reliable 10 dB fade margin above the noise floor at a distance of 80 meters, the transmitter must output an Effective Isotropic Radiated Power (EIRP) of approximately 0 dBm to +2 dBm (1 milliwatt to 1.5 milliwatts). This aligns perfectly with the standard FCC Part 15.249 limits for intentional radiators in the ISM band, meaning the device is tuned for maximum legal transmission power without requiring the complex spread-spectrum techniques mandated by Part 15.247 for higher power outputs.

RF Characteristic	Extracted Specification	Source Documentation
Operating Frequency Range	2420 MHz – 2465 MHz	FCC Test Reports
Modulation Scheme	GFSK	Physical Label / Filings [cite: image 3, 60]
Estimated Channel Bandwidth	1 MHz to 2 MHz	Common proprietary ASIC architecture
Transmission Power (EIRP)	~0 dBm to +2.5 dBm (1 - 1.8 mW)	Derived via 80m range FSPL calculations
Antenna Architecture	PCB Trace (Internal)	Device form factor constraints

5.0 Protocol Stack, Duty Cycle, and Timing Mechanisms

Identifying the logical protocol layer that organizes the raw RF waves into meaningful data is the final theoretical hurdle before practical signal capture. The user query hypothesized the integration of an off-the-shelf protocol stack such as IEEE 802.15.4 (Zigbee/Thread), Bluetooth Low Energy (BLE), or a proprietary sub-GHz protocol.

5.1 Refutation of Standardized Protocol Stacks

The physical evidence instantly invalidates the sub-GHz hypothesis; the device operates exclusively in the 2.4 GHz microwave band [cite: image 3]. Furthermore, a deep architectural analysis firmly rules out the use of BLE or 802.15.4.

Standardized protocols are laden with massive computational and networking overhead. BLE, for instance, requires complex advertising intervals, connection requests, MAC-layer handshakes, and rigid timing slots to maintain a network topology. IEEE 802.15.4 relies on

intricate beaconing and routing tables. In a system explicitly marketed as a "fastest finger first" reactive game buzzer, this overhead is fatal. If 32 buzzers attempt to initiate a standard BLE handshake simultaneously upon a button press, the collision resolution and MAC-layer processing would introduce hundreds of milliseconds of non-deterministic latency. Consequently, the evidence dictates the use of a **highly streamlined, proprietary 2.4 GHz protocol**. Proprietary protocols allow hardware engineers to strip away all standard networking overhead, creating custom packet structures that exist solely to transmit a few bytes of state data with absolute minimal latency.

5.2 Proprietary Packet Framing and Synchronization

In a proprietary GFSK implementation, the packet framing sequence is highly deterministic and heavily optimized for rapid demodulation. A reverse engineer observing the baseband IQ data will likely encounter a packet structure mirroring the following architecture:

1. **Preamble (1-2 Bytes):** A strictly alternating sequence of binary 1s and 0s (typically 0x55 or 0xAA). This preamble serves no data purpose; rather, it allows the host receiver's Automatic Gain Control (AGC) to settle and allows the phase-locked loop (PLL) to synchronize with the incoming clock rate.
2. **Sync Word / Network ID (3-5 Bytes):** A unique identifier acting as the network address. Because educational environments might deploy multiple TM102 host systems in adjacent classrooms without experiencing cross-talk, the Sync Word binds a specific set of 32 buzzers to a specific host. Any packet intercepted without the correct Sync Word is instantly discarded by the hardware receiver, masking it from the MCU.
3. **Payload (1-3 Bytes):** Given the limited operational parameters, the payload is exceptionally small. It only needs to contain the Buzzer Node ID (0x01 through 0x20 to support 32 nodes), the button state (pressed/released), and potentially a battery status flag (as the host can query battery life).
4. **Cyclic Redundancy Check / CRC (1-2 Bytes):** A hardware-calculated checksum appended to the end of the payload to ensure over-the-air data integrity.

5.3 Duty Cycle Regulations and Timing Profiles

Because the proprietary payload is restricted to a handful of bytes, the absolute transmission duration during a single "buzz" event is infinitesimally short. Assuming the standard baseband signaling rate of 1 Mbps or 2 Mbps utilized by 2.4 GHz ASICs, the transmission of a complete 10-byte packet requires merely 80 to 160 microseconds.

This creates an extreme asymmetry in the device's **duty cycle**. The duty cycle is the mathematical ratio of time a device spends actively transmitting RF energy versus time spent idle or receiving. The FCC heavily regulates duty cycles to prevent spectrum monopolization. During standby, the duty cycle of a TM102 responder is effectively 0%. Even during a catastrophic collision event where a buzzer must execute five randomized backoff retries to successfully deliver its packet, the total active transmission time remains well below one millisecond. Over any standard 100-millisecond FCC measurement window, the maximum theoretical duty cycle of this system is less than 1%. This ultra-low duty cycle allows the manufacturer to easily clear Part 15 periodic operation compliance testing while simultaneously delivering the promised 25-hour battery life on standard AAA alkaline cells.

Protocol Parameter	Analytical Deduction	Engineering Rationale
Protocol Stack	Proprietary 2.4 GHz	BLE/802.15.4 overhead introduces unacceptable latency for a reaction game.
Signaling Rate	1 Mbps or 2 Mbps	Standard silicon limits for low-cost GFSK ASICs; optimizes for rapid burst transmission.
Packet Duration	< 200 Microseconds	Payload contains only Node ID and basic state flags.
Active Duty Cycle	< 1% (Burst Periodic)	Required to maintain 25-hour battery life and pass FCC spectrum sharing rules.

6.0 Signal Reverse-Engineering and Hardware Sniffer Methodology

Synthesizing the hardware topology, duplexing mechanics, and PHY layer characteristics establishes a robust methodological framework for practical signal reverse-engineering. The objective of this phase is to isolate the RF energy, capture the raw payload, and decode the proprietary packet structure using an ESP32 microcontroller and an nRF24L01+ transceiver.

6.1 Hardware Sniffer Setup and Channel Sweeping

To capture the transmission, an ESP32 (30-pin DevKit V1) wired to an nRF24L01+ PA+LNA radio transceiver via an adapter base board will be utilized. The ESP32 interfaces with the nRF24L01+ module via standard SPI pins (MOSI, MISO, SCK, CE, and CSN)¹. The PA+LNA adapter base board is essential, as the nRF24 chip draws current spikes that can cause brownouts if powered directly from the ESP32's 3.3V pin.

Because the nRF24L01+ lacks a wideband "waterfall" view and only monitors a single 1 MHz or 2 MHz channel at a time¹, the ESP32 firmware must be programmed to rapidly sweep across the 45 possible channels (2420-2465 MHz)³. The script will pause on each channel briefly to listen for a button press. Once the active channel is detected, it can halt the sweep.

6.2 Promiscuous Mode Exploit and Payload Decoding

Standard nRF24L01+ modules lack a native promiscuous mode; they require a specific 'pipe' or destination address and will discard packets that do not perfectly match⁴. Because the

Retekess proprietary Network ID is unknown, standard receiver scripts will capture nothing. To bypass this, the ESP32 must implement a hardware-level exploit discovered by security researcher Travis Goodspeed⁵. By intentionally misconfiguring the chip's preamble and MAC address checking logic, ambient RF noise tricks the filter, forcing the nRF24L01+ to dump raw on-air GFSK payloads—including the unknown addresses—directly to the ESP32 via SPI⁵. Once the raw hexadecimal data is dumped to the serial monitor, identifying the proprietary packet structure requires differential analysis:

1. **Isolating the Sync Word:** The investigator should capture ten distinct presses from Buzzer 1. By aligning the raw hexadecimal outputs, the bytes at the beginning of the packet that remain static across all ten captures represent the Preamble and the Sync Word (Network ID).
2. **Isolating the Node ID:** The investigator should then capture ten distinct presses from Buzzer 2. Comparing the Buzzer 2 captures against the Buzzer 1 captures will reveal the specific byte index that changed—this isolates the Buzzer Node ID payload.

6.3 Protocol Fuzzing and Host Emulation

Because the analysis confirms the system is half-duplex and state-dependent, a simple replay attack (recording Buzzer 1's transmission and blindly retransmitting it) may not yield full system control. The host maintains complex state machines, particularly during "Rule Mode" and "Random Mode".

True reverse engineering will require host emulation. By analyzing the downlink packets transmitted by the controller when the "Lock" or "Clear" buttons are pressed [cite: image 2], the investigator can decode the command payload structure. Using the same ESP32 and nRF24L01+ hardware (now configured for standard transmission rather than promiscuous sniffing), the investigator can craft synthetic command packets and transmit them to the edge nodes, effectively hijacking the visual LED feedback mechanisms and overriding the proprietary arbitration logic.

7.0 Conclusion and Final Engineering Assessment

The Retekess TM101/TM102 wireless responder system (FCC ID 2A3NOTM101) represents a highly optimized, application-specific wireless network. Through comprehensive analysis of the FCC OET exhibits, physical hardware labeling, and operational paradigms, the system's underlying architecture has been fully demystified.

The network operates on a strict point-to-multipoint topology, completely eschewing synchronous polling in favor of an asynchronous, interrupt-driven architecture to guarantee microsecond precision during "fastest finger first" arbitration. The necessity for dynamic, host-driven visual feedback to the players confirms that the remote edge nodes are not simplex devices, but fully capable half-duplex transceivers.

At the physical layer, the system leverages GFSK modulation within a firmware-restricted 2420 MHz to 2465 MHz window, avoiding the extremes of the 2.4 GHz ISM band to ensure regulatory compliance. The strict latency requirements definitively rule out the use of heavy standardized protocols like BLE or 802.15.4, pointing entirely to a lightweight, low-duty-cycle proprietary protocol. For the security researcher or reverse engineer, successfully exploiting this system relies on implementing hardware-level promiscuous mode exploits on an nRF24L01+ sniffer and

utilizing differential baseband analysis to isolate the static network sync words from the dynamic node identifiers.

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